Python Coding Series # 3

Introduction to Bayesian Modeling with PyMC

Jeffrey Covington

May 12, 2026

Part I

Introduction to Bayesian Statistics

What Are We Going to Talk About?

This workshop will cover all of the following:

- Bayesian concepts and philosophy
- Algorithms for fitting Bayesian models
- Software for Bayesian Statistics
- Practical use of Bayesian Statistics

What is Bayesian Statistics?

Bayesian Statistics ...

- ...is a powerful and flexible statistical modeling framework.
- ...is possible in the 21st century due to advances in computing power.
- ...is the cutting edge of new statistical methods.

The Central Idea of Bayesian Thinking

- Probability represents degree of belief, i.e. uncertainty.
- All uncertainty can be represented as probability.

  ... at least in theory.

Key Differences Between Bayesian and Classical Statistics

Classical Statistics

- Outputs a single "point estimate" of a parameter.
- Many specialized tools for specific situations.
- Implicit assumptions baked into the tools.

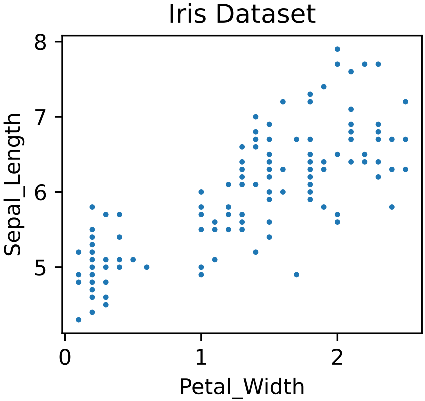
Bayesian Statistics

- Outputs a distribution of plausible paramter values.
- General tools for any situation.
- You need to make your assumptions explicit.

Case Study: Linear Regression

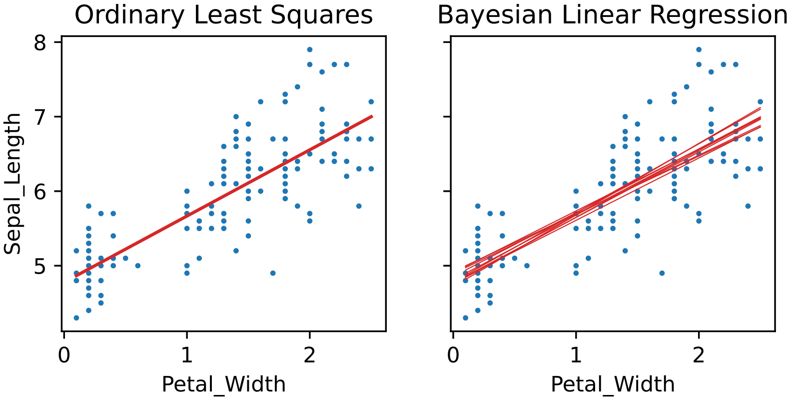
Let's compare classical and Bayesian statistics on a simple dataset.

```
     Petal_Width  Sepal_Length
0            0.2           5.1
1            0.2           4.9
2            0.2           4.7
3            0.2           4.6
4            0.2           5.0
..           ...           ...
145          2.3           6.7
146          1.9           6.3
147          2.0           6.5
148          2.3           6.2
149          1.8           5.9

[150 rows x 2 columns]
```



One Regression Line vs. Many Lines

- Ordinary least squares produces a single regression line.
- Bayesian regression produces a distribution of plausible regression lines.



Probabilities as Uncertainties

- In Bayesian Statistics, all unknown quantities have an associated probability (distribution.)
    - For a true/false proposition, this is a probability.
    - For an unknown quantity/parameter, this is a probability distribution.
- We don't need uncertainty about uncertainty. It can all be collected into a single probability distribution.

What is a Conditional Probability?

- A conditional probability is the probability of some event **given** we know some piece of information.
- For example, compare these two:
  - What is the probability it will rain in 1 hour?
  - What is the probability it will rain in 1 hour given it is overcast?

The Prior and Posterior Distributions

- The **prior** probability distribution of a quantity is estimate of a quantity **before** collecting data.
- The **posterior** probability distribution is our estimate of a quantity **after** collecting data.
- For example, consider estimating the mean height of a population.
  - The prior estimate could be taken from prior research on similar populations.
  - The posterior estimate could be taken from a sample mean of a representative sample.

Breaking Down Bayes' Theorem

- Bayesian Statistics is named after Bayes' Theorem from Probability Theory.
- Bayes' Theorem tells us how to calculate the posterior distribution (of parameters.)

$$\overbrace{\underset{\substack{\uparrow \\ \text{“probability of”}}}{P}(\text{parameters} \underset{\substack{\uparrow \\ \text{“given”}}}{\mid} \text{data})}^{\text{Posterior}} = \frac{\overbrace{P(\text{data} \mid \text{parameters})}^{\text{Model}} \times \overbrace{P(\text{parameters})}^{\text{Prior}}}{\underbrace{P(\text{data})}_{\text{Normalizing factor}}}$$

Note: The denominator is complicated but straightforward to calculate.

Key Takeaways from Bayes' Theorem

- Bayes' Theorem gives us a precise definition of statistical evidence.
- We must quantify uncertainty.
- Data is only one part of the picture.
- Your prior beliefs matter, whether explicitly or through implicit assumptions.

Comparing Classical and Bayesian Statistics

- All Statistical frameworks deal with statistical models and how to fit their parameters.
- Classical and Bayesian Statistics agree on the math. Both are correct.
- Classical and Bayesian Statistics differ in:
  - How to fit models.
  - Priors and (unstated) assumptions.
  - Convenience and computational cost.

Which is Better?

Which approach, classical (frequentist) or Bayesian, is better?

- Nether approach disagree about the mathematics.
- If you have a lot of data, the approach doesn't really matter. The data will speak for itself.

When Should You Use Bayesian Statistics?

Here are examples where you should consider Bayesian Statistics:

- You have small sample sizes.
- You have a custom, complex model.
- You want to incorporate prior knowledge.

Comparing Summary Statistics of Classical and Bayesian Approaches

Classical Statistics and Bayesian Statistics have subtly different terminology.

- Hypothesis testing is less prevalent in Bayesian Statistics. When it is used, the "Bayes factor" is used intead of a p-value.
- The analog of confidence intervals are credible intervals.
- One can come up directly with analogues to many concepts in classical statistics (p-values, maximum likelihood estimates, etc.) if needed.

Part II

Bayesian Modeling in Practice

How Do You Build a Bayesian Model?

- A model describes a mathematical relationship between the response variables and explanatory variables.
- Historically, fitting the model parameters is the hard part.
- Modern software packages make this much more accessible.

How Do You Fit a Bayesian Model?

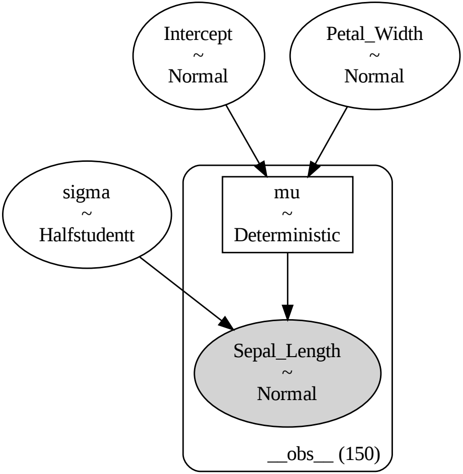
- Markov Chain Monte Carlo (MCMC) is the key algorithm which makes Bayesian Statistics possible.
- MCMC tries many configurations of parameters, a sophisticated "guess and check" algorithm.
- MCMC is compuatationally expensive, which is why Bayesian modeling has only become popular in the 21st century.

How Do You Build a Bayesian Model?

- Because Markov Chain Monte Carlo is so powerful, Bayesian Models can be very flexible.
- There are software packages to build and fit Bayesian Models.
  - Packages like Bambi have built-in Bayesian models. (Highest convenience.)
  - Packages like PyMC let you build models in a structured way. (Balanced approach.)
  - Packages like PyTensor give you maximum flexibility but demand a lot from you.

Visualizing Bayesian Model Structure with a Graph

- Bayesian Models are usually illustrated with a directed graph.
- Nodes are the random variables of the model.
- It shows the prior for each variable.
- Arrows represent dependencies.
- It does **not** show how variables are realted.



How Do You Choose Your Priors?

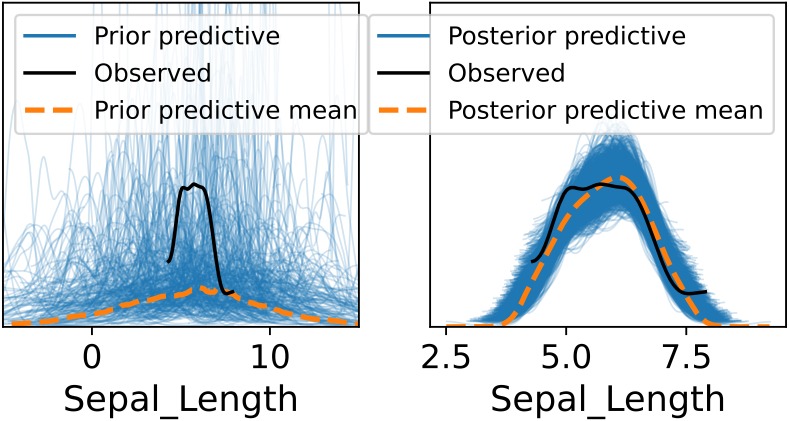
- You can choose uninformative priors when you want to make as few assumptions as possible. This is the most common use case.
- When you have a small sample size, you need to consider your priors carefully.
- Priors can be chosen from domain or expert knowledge.
- Pilot studies can be very useful for choosing priors.

What distributions can you choose?

- Gaussian distributions are common, but you can choose many other distributions.
- "Fat-tailed" distributions (Student's t-distribution, Cauchy, etc.)
- Multi-modal distributions.
- Distributions with finite support (Uniform, Beta, etc.)
- Discrete distributions (Poisson, Negative Binomial, etc.)

Prior and Posterior Predictive Checks

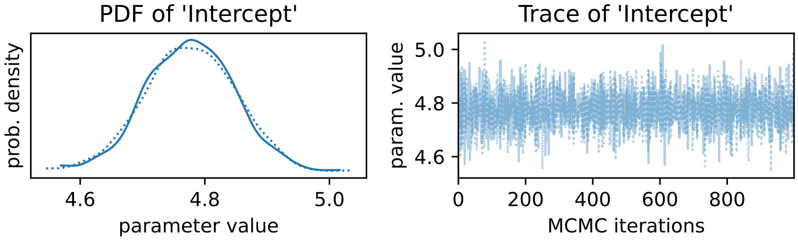
- We can check the “predicted” data from both the prior and posterior.
- These plots tell us:
  - Whether our priors are reasonable.
  - Whether our fit is good.



Considerations of Markov Chain Monte Carlo

- Most Bayesian Statistical methods are based on Markov Chain Monte Carlo (MCMC).
- MCMC is stochastic (random) and has several underlying assumptions.
- We want to see:
  - The MCMC process converge (to the posterior distribution.)
  - Low serial correlation.

Interpreting Trace Plots



- On the left is the probability density function.
- On the right is the trace plot.
  - It is stable (it has converged.)
  - It looks very “fuzzy”, meaning low serial correlation.

Bayesian Model Output

Here is an example of Bayesian fitting output:

```
             mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd
Intercept   4.776  0.071   4.645    4.909      0.001    0.001
Petal_Width 0.890  0.051   0.783    0.978      0.001    0.001
sigma       0.481  0.029   0.431    0.534      0.001    0.001
```

mean The mean of the posterior.

sd The standard deviation of the posterior.

hdi_3% The lower bound of the 95% credible interval.

hdi_97% The upper bound of the 95% credible interval.

mcse_mean The mean of the Monte Carlo Standard Error. Should be close to 0.

mcse_sd The standard deviation of the Monte Carlo Standard Error.

Bayesian Model Output

Here is an example of Bayesian fitting output.

```
             ess_bulk  ess_tail  r_hat
Intercept      3706.0    1721.0    1.0
Petal_Width    3226.0    1664.0    1.0
sigma          2983.0    1204.0    1.0
```

ess_bulk The effective sample size within the 95% credible interval (the "bulk".)

ess_tail The effective sample size outside the 95% credible interval (the "tail".)

r_hat $\hat{R}$, a measure of MCMC convergence. Should be close to 1.0.

Part III

Applied Bayesian Modeling Using PyMC

What is PyMC?

- PyMC is a Bayesian modeling framework in Python.
- Bambi is built on top of PyMC and has pre-built Bayesian models and uses R-style formulae.
  - For typical statistical analyses, Bambi is very convenient.
  - For more flexibility and complexity, PyMC is needed.
- ArviZ is used to create plots and tables for Bayesian Statistics.

Bayesian Workflow in Python

1. Import and preprocess data with Pandas.
2. Build and fit your model with Bambi or PyMC
3. Create and export figures and tables with ArviZ.

Using Bambi for Bayesian Modeling

Here is how to build and fit a model with Bambi.

Input:

```
import bambi as bmb
import arviz as az
model = bmb.Model("Petal_Length ~ Sepal_Width",
                  data=data)
result = model.fit()
print(az.summary(bmb_result))
```

Output:

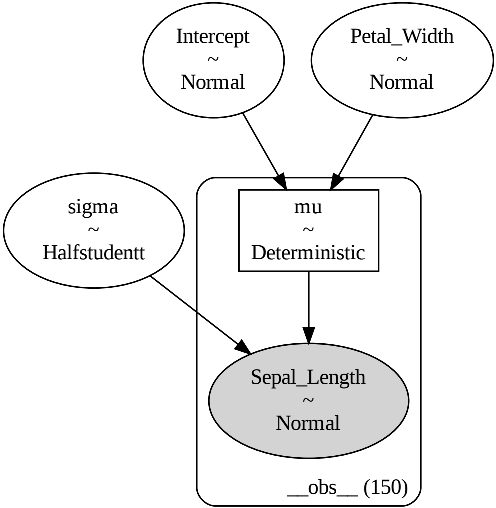
```
              mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
Intercept    4.776  0.071   4.645    4.909      0.001    0.001    3706.0
Petal_Width  0.890  0.051   0.783    0.978      0.001    0.001    3226.0
sigma        0.481  0.029   0.431    0.534      0.001    0.001    2983.0

             ess_tail  r_hat
Intercept      1721.0    1.0
Petal_Width    1664.0    1.0
sigma          1204.0    1.0
```

Visualizing the Graph of the Model

Input:

```
model.graph()
```

Output:

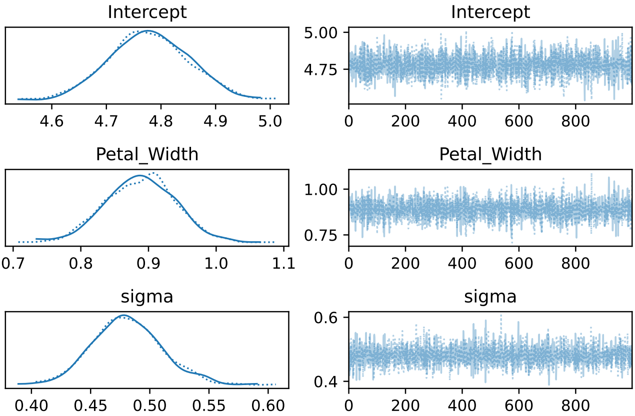


Creating the Trace Plot

Input:

```
az.plot_trace(result)
```

Output:



Exporting Figures in Python

- Almost all figures in Python, including those produced by ArviZ, are based on matplotlib.
- Figures produced by different packages are interoperable using matplotlib.
- You use matplotlib to export figures.

Building Custom Models with PyMC

- PyMC allows you to fully customize your model.
- Models are built up one variable at a time, forming a tree.

Building Custom Models with PyMC

Here is how to create the same model with PyMC.

Input:

```
import pymc as pm
with pm.Model() as pm_model:
    x = data["Sepal_Width"]
    y = data["Petal_Length"]
    sigma = pm.HalfStudentT("sigma", nu=4, sigma=10)
    intercept = pm.Normal("Intercept", 0, sigma=10)
    slope = pm.Normal("slope", 0, sigma=10)
    mu = pm.Deterministic("mu", intercept + slope * x)
    likelihood = pm.Normal("y", mu=mu, sigma=sigma,
                           observed=y)
```

Unit 3 Lab: Hands-On with Bambi and PyMC

Please open the Unit 3 Lab on the GitHub page.

Part IV

Reporting Bayesian Modeling Results

Case Study: Examining a Paper

We are going to examine a paper which uses Bayesian Statistics:

Corponi, F., Li, B.M., Anmella, G. *et al.* A Bayesian analysis of heart rate variability changes over acute episodes of bipolar disorder. *npj Mental Health Res* **3**, 44 (2024). https://doi.org/10.1038/s44184-024-00090-x

Case Study: Examining a Paper

We are going to examine a paper which uses Bayesian Statistics:

Corponi, F., Li, B.M., Anmella, G. *et al.* A Bayesian analysis of heart rate variability changes over acute episodes of bipolar disorder. *npj Mental Health Res* **3**, 44 (2024). https://doi.org/10.1038/s44184-024-00090-x

Things to Look Out For

- What tables and figures did they decide to include or exclude?
- Why did they choose to do Bayesian Analysis?
- How do they address the limitations of their analysis?

Unit 4 Lab: Reporting Bayesian Modeling Results

Please open the Unit 4 Lab on the GitHub page.

Thank you!

Please fill out the survey:

`https://tinyurl.com/RCCWorkshop-Covington3`